usepackage[a4paper,margin=0.65in]geometry

812.

ON ARCHIMEDES' THEOREM FOR THE SURFACE OF A CYLINDER.

[From the *Messenger of Mathematics*, vol. xiii. (1884), pp. 107, 108.]

The measure of the surface of a cylinder was first obtained by Archimedes in his Treatise on the Sphere and Cylinder (Book i., Prop. xiv.), *Œuvres d'Archimède*, par F. Peyrard, 4° Paris, 1807, pp. 26–31; viz. Archimedes showed that the surface of the cylinder was equal to the area of a circle, having its radius a mean proportional between the height and the diameter of the circular base [$S = 2\pi ah, = \pi\{\sqrt{(2a \,.\, h)}\}^2$].

The following is in effect his demonstration:

He considers regular polygons (with the same number of sides) inscribed in and circumscribed about a circle; and, as regards the cylinder, the prisms standing on these polygons.

Say for the circular base of the cylinder we have

$S^\times$	surface of circumscribed prism,
S	cylinder,
$S_\times$	inscribed prism;

and for the circle, having its radius a mean proportional between the height and the diameter of the circular base,

$B^\times$	area of circumscribed polygon,
B	circle,
$B_\times$	inscribed polygon,

where the four polygons referred to by $S^\times$, $S_\times$, $B^\times$, $B_\times$ have all of them the same number of sides.

It is in the preceding propositions (by means of an axiom as to curve lines) shown that

$$S^\times > S > S_\times, \quad B^\times > B > B_\times\,;$$

and it is further shown that

$$S^\times = B^\times, \quad S_\times = B_\times.$$

It is moreover shown that, by taking the number of sides sufficiently large, the ratio $B^\times : B_\times$, or say the fraction $B^\times/B_\times$ (which is greater than 1) may be made less than any given quantity $1 + \varepsilon$.

It is then to be shown that $S = B$.

If not, then

either

$B < S$.

This being so, it is possible to make

$B_\times/B^\times < S/B$,

that is,

$S^\times/B^\times < S/B$,

or

$S^\times/S < B^\times/B$,

which is absurd, since

$S^\times/S > 1\,;\ B^\times/B < 1\,;$

or else

$B > S$.

This being so, it is possible to make

$B_\times/B^\times < B/S$,

that is,

$B^\times/S^\times < B/S$,

or

$B^\times/B < S^\times/S$,

which is absurd, since

$B^\times/B > 1\,;\ S^\times/S < 1\,;$

and consequently $S = B$, the theorem in question.

I take the opportunity of referring to two theorems by Archimedes, Lemmas, Prop. v. and vi., Peyrard, pp. 429–435, which relate to the contacts of circles. We have in each of them the figure which he calls the Arbelon, viz. if A, C, B are points in this order on the same straight line, then the figure consists of the three semicircles on the diameters AC, CB, and AB respectively, and the Arbelon is the space included between the three semi-circumferences.

In Prop. v., we have the common tangent at C to the two semicircles AC, CB; this divides the Arbelon into two mixtilinear triangles (each bounded by the common tangent, one of the smaller semicircles, and a portion of the larger semi-circle), and inscribing each of these a circle, the theorem is that the two inscribed circles are of equal magnitude.

In Prop. vi., the theorem is that the radii of the smaller semicircles being as 3 : 2, then the radius of the circle inscribed in the Arbelon is to the diameter of the larger semicircle as 6 to 19. But it is noticed that the demonstration would apply to any other value of the ratio; and, in fact, if the radii of the two smaller circles are as $a : b$, then the radius of the inscribed circle is to the diameter of the larger semicircle as ab to $a^2 + ab + b^2$, which is the general form of the theorem.

813.

[NOTE ON MR GRIFFITHS' PAPER "ON A DEDUCTION FROM THE ELLIPTIC-INTEGRAL FORMULA $y = \sin(A + B + C + \ldots)$".]

[From the *Proceedings of the London Mathematical Society*, vol. xv. (1884), p. 81.]

Consider, for instance,

the cubic transformation

$$y = \frac{x\,[1 + 2\alpha' - (1+\alpha')^2\,x^2]}{1 - \alpha^2 x^2},$$

where $\alpha^2 + \alpha'^2 = 1$.

This implies

$$\sqrt{1-y^2} = \frac{\sqrt{1-x^2}\,[1 - (1+\alpha')^2\,x^2]}{1-\alpha^2x^2},$$

viz., $\sqrt{1-y^2}$ = a rational multiple of $\sqrt{1-x^2}$.

Also the quadric transformation

$$z = \frac{1 - (1+\beta'^2)\,x^2}{1-\beta^2x^2}\,;$$

where $\beta^2 + \beta'^2 = 1$.

This implies

$$\sqrt{1-z^2} = \frac{\sqrt{1-x^2}\,.\,2\beta' x}{1-\beta^2x^2},$$

viz., $\sqrt{1-z^2}$ = a rational multiple of $\sqrt{1-x^2}$.

Hence, assuming

$$u = yz - \sqrt{1-y^2}\,\sqrt{1-z^2},$$

which is a rational function

$$= \frac{x\,(a_0 - a_2x^2 + a_4x^4)}{1-\alpha^2x^2\,.\,1-\beta^2y^2},$$

we have

$$\sqrt{1-u^2} = y\sqrt{1-z^2} + z\sqrt{1-y^2},$$

which is $= \sqrt{1-x^2}$ multiplied by a like rational function.

That is, in defining the a_0, a_2, a_4 functions of the two arbitrary coefficients α, β, as above, we have in effect so determined them that $\sqrt{1-u^2}$ shall be $=\sqrt{1-x^2}$ multiplied by a rational function of x.

We can then further determine a_0, a_2, a_4 in such wise that the change of x into $\dfrac{1}{kx}$ shall change u into $\dfrac{1}{\lambda u}$; and, this being so, making the change in $\sqrt{1-x^2}$, we obtain $\sqrt{1-\lambda^2 u^2}$ in the form, $\sqrt{1-k^2x^2}$ multiplied by a rational function of x; viz. u is a function of x such that

$$\frac{du}{\sqrt{1-u^2}\,\sqrt{1-\lambda^2u^2}} = \frac{M\,dx}{\sqrt{1-x^2}\,\sqrt{1-k^2x^2}}.$$

The theory is thus in effect Jacobi's—with the novelty of combining two lower transformations in such wise that the assumed expression for u as a rational function of x shall give

$$\sqrt{1-u^2} = \sqrt{1-x^2} \text{ multiplied by a rational function of } x.$$

It is not necessary that the equations

$$y = \text{rational function of } x \quad \text{and} \quad z = \text{rational function of } x$$

should be elliptic-function transformations. All that is required is that they should be such as to give $\sqrt{1-y^2}$ and $\sqrt{1-z^2}$ each $=\sqrt{1-x^2}$ multiplied by a rational function of x.

8–2

814.

ON DOUBLE ALGEBRA.

[From the *Proceedings of the London Mathematical Society*, vol. xv. (1884), pp. 185–197. Read April 3, 1884.]

1. I consider the Double Algebra formed with the extraordinary symbols, or "extraordinaries" x, y, which are such that

$$\begin{aligned} x^2 &= ax + by, \\ xy &= cx + dy, \\ yx &= ex + fy, \\ y^2 &= gx + hy, \end{aligned}$$

or, as these equations may also be written,

$$\begin{array}{c|cc} & x & y \\ \hline x & (a,\ b) & (c,\ d) \\ y & (e,\ f) & (g,\ h) \end{array},$$

where a, b, c, d, e, f, g, h are ordinary symbols, or say coefficients; all coefficients being commutative and associative *inter se* and with the extraordinaries x, y.

The system depends in the first instance on the eight parameters a, b, c, d, e, f, g, h; but we may, instead of the extraordinaries x, y, consider the new extraordinaries connected therewith by the linear relations $\xi = \alpha x + \beta y$, $\eta = \gamma x + \delta y$, where the coefficients α, β, γ, δ may be determined so as to establish between the eight parameters any four relations at pleasure (or, what is the same thing, α, β, γ, δ are what I call "apocatastic" constants): and the number of parameters is thus properly $8 - 4, = 4$.

2. The extraordinaries here considered are not in general associative; differing herein from the imaginaries of Peirce's Memoir, "Linear Associative Algebra" (1870),

reprinted in the *American Mathematical Journal*, t. iv. (1881), pp. 97–227, which as appears by the title, refers only to associative imaginaries. I recall some definitions and results. The symbol x is said to be *idempotent* if $x^2 = x$, *nilpotent* if $x^2 = 0$; and the systems of associative symbols are expressed as much as may be by means of such idempotent and nilpotent symbols: thus the linear systems are $(a_1)\, x^2 = x$, $(b_1)\, x^2 = 0$. A double linear system composed of independent symbols, that is, symbols x, y each belonging to its own linear system, and moreover such that $xy = yx = 0$, is said to be "mixed"; thus the mixed double systems are

$$\begin{array}{c|cc} & x & y \\ \hline x & x & 0 \\ y & 0 & y \end{array}\,, \qquad \begin{array}{c|cc} & x & y \\ \hline x & x & 0 \\ y & 0 & 0 \end{array}\,, \qquad \begin{array}{c|cc} & x & y \\ \hline x & 0 & 0 \\ y & 0 & 0 \end{array}\,.$$

But Peirce excludes those from consideration, attending only to the pure systems, which he finds to be

$$(a_2)\ \begin{array}{c|cc} & x & y \\ \hline x & x & y \\ y & y & 0 \end{array}\,, \qquad (b_2)\ \begin{array}{c|cc} & x & y \\ \hline x & x & y \\ y & 0 & 0 \end{array}\,, \qquad (c_2)\ \begin{array}{c|cc} & x & y \\ \hline x & 0 & x \\ y & 0 & 0 \end{array}\,.$$

To these, however, should be added the system

$$(d_2)\ \begin{array}{c|cc} & x & y \\ \hline x & x & 0 \\ y & y & 0 \end{array}\,;$$

see post, No. 19.

3. In the general theory, where the symbols are not in the first instance taken to be associative, we may of course establish between the coefficients such relations as will make the symbols associative; and the question presents itself to show how .in this case the system reduces itself to one of Peirce's systems. This I considered in my note "On Associative Imaginaries," *Johns Hopkins University Circular*, No. 15 (1882), p. 211 [822]; I there obtained, as the general form of the commutative and associative system,

$$\begin{aligned} x^2 &= ax + by, \\ xy = yx &= cx + dy, \\ y^2 &= \frac{cd}{b}\,x + \frac{d^2 + bc - cd}{b}\,y, \end{aligned}$$

the relation of which to Peirce's system was, as I there remarked, pointed out to me by Mr C. S. Peirce: this will be considered in the sequel, Nos. 13 to 19.

4. Starting now with the general equations

$$\begin{aligned} x^2 &= ax + by, \\ xy &= cx + dy, \\ yx &= ex + fy, \\ y^2 &= gx + hy, \end{aligned}$$

we may attempt to find an extraordinary; $\xi, = \alpha x + \beta y$ (α, β coefficients), such that $\xi^2 = K\xi$ (K a coefficient). In general, K is not $= 0$, and when it is not $= 0$, it may without loss of generality be taken to be $= 1$; we have then $\xi^2 = \xi$, the *idempotent* symbol. But K may be $= 0$; and then $\xi^2 = 0$, ξ a *nilpotent* symbol. To include the two cases, I retain K, it being understood that, when K is not $= 0$, it may be taken to be $= 1$. We have

$$\begin{aligned} \xi^2 &= \alpha^2\,(ax + by) + \alpha\beta\,(\overline{c+e}\,x + \overline{d+f}\,y) + \beta^2\,(gx + hy) \\ &= [\alpha^2 a + (c+e)\,\alpha\beta + g\beta^2]\,x + [\alpha^2 b + (d+f)\,\alpha\beta + h\beta^2]\,y. \end{aligned}$$

Hence, when this is $= K\xi$, we have

$$\frac{\alpha}{\beta} \cdot K(\alpha x + \beta y),$$

$$\frac{\alpha}{\beta} = \frac{\alpha^2 a + (c+e)\,\alpha\beta + g\beta^2}{\alpha^2 b + (d+f)\,\alpha\beta + h\beta^2},$$

a cubic equation for the determination of the ratio $\alpha : \beta$; and, for any particular value of the ratio, we can in general determine the absolute magnitudes, so that

$$K, = \frac{1}{\alpha}\,[\alpha^2 a + (c+e)\,\alpha\beta + g\beta^2], = \frac{1}{\beta}\,[b\alpha^2 + (d+f)\,\alpha\beta + h\beta^2],$$

shall be $= 1$. If, however, for the given value of the ratio we have only

$$\alpha^2 a + (c+e)\,\alpha\beta + g\beta^2 = 0, \quad b\alpha^2 + (d+f)\,\alpha\beta + h\beta^2 = 0,$$

one of these equations, of course, implying the other, then the value of K is $= 0$.

5. It follows that there are in general three idempotent symbols ξ, η, ζ, that is, extraordinaries such that $\xi^2 = \xi$, $\eta^2 = \eta$, $\zeta^2 = \zeta$. The cubic equation may, however, have two equal roots, or three equal roots identically; in this last case, any extraordinary $\alpha x + \beta y$ is an idempotent symbol, and we have to consider in detail further on we may, instead of an idempotent symbol or

symbols, have a nilpotent symbol or symbols. It might be convenient to use the term Pierce's symbol which is in general idempotent, but which may be also nilpotent. Writing $\dfrac{\alpha}{\beta} = \dfrac{-y}{x}$, we obtain a cubic equation $\Omega = (x, y)^3 = 0$, where obviously the linear factors of Ω are the just-mentioned functions ξ, η, ζ; that is, we have ξ, η, ζ as the linear factors of the cubic function

$$\Omega = gx^3 + (h - c - e)\, x^2 y + (a - d - f)\, xy^2 + by^3;$$

each such factor, except in the case where it is nilpotent, being determined so that it shall be idempotent. The cubic function of course vanishes identically if

$$g = 0, \quad h - c - e = 0, \quad a - d - f = 0, \quad b = 0.$$

6. Two extraordinaries $\xi, = \alpha x + \beta y$; $\eta, = \gamma x + \delta y$ (α, β, γ, δ coefficients); may be such that $\xi\eta = 0$; this, of course, does not imply $\eta\xi = 0$; we have, in fact,

$$\begin{aligned}\xi\eta &= (\alpha x + \beta y)\,(\gamma x + \delta y)\\ &= \alpha\gamma x^2 + \alpha\delta xy + \beta\gamma yx + \beta\delta y^2\\ &= \alpha\gamma\,(ax + by) + \alpha\delta\,(cx + dy) + \beta\gamma\,(ex + fy) + \beta\delta\,(gx + hy)\\ &= (a\alpha\gamma + c\alpha\delta + e\beta\gamma + g\beta\delta)\,x + (b\alpha\gamma + d\alpha\delta + f\beta\gamma + h\beta\delta)\,y\,;\end{aligned}$$

and the required condition is satisfied if

$$a\alpha\gamma + c\alpha\delta + e\beta\gamma + g\beta\delta = 0,$$

$$b\alpha\gamma + d\alpha\delta + f\beta\gamma + h\beta\delta = 0.$$

Writing these equations first under the form

$$\gamma\,(a\alpha + e\beta) + \delta\,(c\alpha + g\beta) = 0,$$

$$\gamma\,(b\alpha + f\beta) + \delta\,(d\alpha + h\beta) = 0,$$

and then under the form

$$\alpha\,(a\gamma + c\delta) + \beta\,(e\gamma + g\delta) = 0,$$

$$\alpha\,(b\gamma + d\delta) + \beta\,(f\gamma + h\delta) = 0,$$

we have

$$(a\alpha + e\beta)\,(d\alpha + h\beta) - (b\alpha + f\beta)\,(c\alpha + g\beta) = 0,$$

a quadric equation for the determination of $\alpha : \beta$; and similarly

$$(a\gamma + c\delta)\,(f\gamma + h\delta) - (b\gamma + d\delta)\,(e\gamma + g\delta) = 0,$$

a quadric equation for the determination of $\gamma : \delta$; that is, there are two values of the left-hand factor ξ, and two values of the right-hand factor η. But, of course, these correspond each to each, viz. either factor being given, the other factor is determined uniquely.

Writing successively $\dfrac{\alpha}{\beta} = \dfrac{-y}{x}$, and $\dfrac{\gamma}{\delta} = \dfrac{-y'}{x'}$, we have the quadric functions $(x, y)^2$,

$$\Phi = (ah - fg)\,x^2 + (-ah - cd + bg + cf)\,xy + (cd - be)\,y^2,$$

$$\Phi_1 = (ah - dg)\, x'^2 + (-ah - cf + bg + de)\, x'y' + (cf - be)\, y'^2,$$

where the linear factors of Φ are the two values ξ_1, ξ_2 of the left-hand factor ξ, and the linear factors of Φ_1 are the two values η_1, η_2 of the right-hand factor η.

7. In the commutative case, $c = e$, and $d = f$, we have

$$\Phi = \Phi_1 = (ah - dg)\, x^2 + (-ah + bg)\, xy + (cd - be)\, y^2\, ;$$

here $\xi\eta = 0$, $\eta\xi = 0$, and the values may be taken to be $(\xi_1,\ \eta_1) = (\xi,\ \eta)$, $(\xi_2,\ \eta_2) = (\eta,\ \xi)$, so that $\xi_1\xi_2 = \xi\eta$, as above. The value of Ω is

$$\Omega = gx^3 + (h - 2c)\,x^2y + (a - 2d)\,xy^2 + by^3.$$

8. In the commutative and associative case, taking ξ, η, ζ to be the three idempotent symbols, $\xi^2 = \xi$, $\eta^2 = \eta$, $\zeta^2 = \zeta$; we have

$$\xi\,(\eta - \xi\eta) = \xi\eta - \xi^2\eta = \xi\eta - \xi\eta = 0\,;$$

and in like manner we have the six equations

$$\xi\,(\eta{-}\xi\eta) = 0,\ \xi\,(\zeta{-}\xi\zeta) = 0\,;\ \eta\,(\xi{-}\eta\xi) = 0,\ \eta\,(\zeta{-}\eta\zeta) = 0\,;\ \zeta\,(\xi{-}\zeta\xi) = 0,\ \zeta\,(\eta{-}\zeta\eta) = 0\,;$$

viz. regarding the right-hand factor as being in each case expressed as a linear function of x, y, we have apparently six distinct factors $\eta - \xi\eta$, $\zeta - \xi\zeta$, &c. each of these products would be in any one case identically $= 0$, viz. the case of the second factor being $= 0$. But we know there are only two right-hand factors η_1, η_2, that is, $\eta - \xi\eta$ must be in some way or other identically a linear function of ξ; the same as $\zeta - \xi\zeta$, but here suppose (a, b coefficients, neither of them $= 0$); we thence have

$$\xi\,(a\xi + b\eta) = a\xi^2 + b\xi\eta,\ = a\xi + b\eta,\ = a\xi + b\eta,$$

that is, $(a^2 - a)\,\xi + (b^2 - b)\,\eta = 0$; whence $a = 1$, $b = 1$, and therefore $\xi\eta = \xi + \eta$; hence also $\xi\zeta$ and $\eta\zeta = \xi + \zeta$ and $\eta + \zeta$; $\xi\eta\,\eta\zeta$. The six products consequently are $\xi\eta$, $\eta\zeta$, $\eta\xi$, $\zeta\xi$, $\zeta\eta$, each $= 0$ or identically $= 0$.

9. In verification of the theorem that for the commutative and associative system the cubic function Ω contains the quadric function Φ as a factor, we may write, as above,

$$y = \frac{cd}{b}, \quad h = \frac{d^2 + bc - cd}{b},$$

values which give

$$\begin{aligned} b\Phi &= (bc^2 - acd)\,x^2 + (-ad^2 - abc + a\,d + bcd)\,xy + (bcd - b^2e)\,y^2 \\ &= -(ad - bc)\,[cx^2 + (d - a)\,xy - by^2], \\ b\Omega &= cdx^3 + (d^2 - bc)\,x^2y + (ab - 2bd)\,xy^2 + b^2y^3, \\ &= (dx - by)\,[cx^2 + (d - a)\,xy - by^2], \end{aligned}$$

which gives the theorem in question. And observe further that

$$\begin{aligned} (dx - by)\xi = d^2\,(ax + by)\,c - 2bd\,(cx + dy) + b^2\,\left(\frac{cd}{b}\,x + \frac{d^2 + bc - cd}{b}\,y\right), \\ = -(ad - bc)\,(dx - by), \end{aligned}$$

that is, disregarding coefficients, the two idempotent symbols ξ, η are the linear factors of $cx^2 + (d - a)\,xy - by^2$, and the third idempotent symbol ζ is $= dx - by$.

10. Introducing coefficients in order to make the symbols ξ, η, ζ idempotent, and writing accordingly

$$\xi = \frac{1}{K}\,[cx + \tfrac{1}{2}\,(d - a + \sqrt{\nabla}\,)\,y], \quad \nabla = (d-a)^2 + 4bc,$$

so that

$$\eta = \frac{1}{L}\,[cx + \tfrac{1}{2}\,(d - a - \sqrt{\nabla}\,)\,y], \quad \xi\eta = \frac{c}{KL}\,[cx^2 + (d-a)\,xy - by^2],$$

$$\zeta = \frac{1}{P}\,(dx - by),$$

we have to verify that it is possible to determine K, L, P so that $\xi^2 = \xi$, $\eta^2 = \eta$, $\zeta^2 = \zeta$, $\xi\eta = \xi + \eta$. The last equation gives

$$\frac{d}{P} = \frac{c}{K} + \frac{c}{L},$$

$$-\frac{2b}{P} = \frac{d - a + \sqrt{\nabla}}{K} + \frac{d - a - \sqrt{\nabla}}{L},$$

and we thence have

$$\frac{d\,(d-a) + 2bc - d\sqrt{\nabla}}{P} = -\frac{2c\sqrt{\nabla}}{K},$$

$$\frac{d\,(d-a) + 2bc + d\sqrt{\nabla}}{P} = \frac{2c\sqrt{\nabla}}{L},$$

and we can from the equation $\xi^2 = \xi$ find P; viz. comparing the coefficients of x, we have

$$\frac{d}{P} = \frac{1}{P^2}\left(ad^2 - 2b\,dc + b\frac{cd}{b}\right),\ = \frac{d}{P^2}\,(ad - bc), \text{ that is, } P = (ad - bc),$$

or the values of K and L are

$$\frac{[d\,(d-a) + 2bc - d\sqrt{\nabla}\,]}{ad - bc} = -\frac{2c\sqrt{\nabla}}{K},$$

$$\frac{[d\,(d-a) + 2bc + d\sqrt{\nabla}\,]}{ad - bc} = +\frac{2c\sqrt{\nabla}}{L},$$

which should agree with the values of K and L found from the equations $\xi^2 = \xi$, $\eta^2 = \eta$, respectively. Comparing the coefficients of x, the first of these equations gives

$$\begin{aligned}
\frac{c}{K} &= \frac{1}{K^2}\left\{c^2a + c\,(d - a + \sqrt{\nabla}\,)\,c + \tfrac{1}{4}\,(d - a + \sqrt{\nabla}\,)^2\,\frac{cd}{b}\right\} \\
&= \frac{c}{4bK^2}\left\{4abc + 4bc\,(d-a) + d\,[(d-a)^2 + \nabla\,] + 2\,[2bc + d\,(d-a)]\,\sqrt{\nabla}\right\} \\
&= \frac{c}{2bK^2}\left\{d\nabla + [d\,(d-a) + 2bc]\,\sqrt{\nabla}\right\},
\end{aligned}$$

that is,

$$K = \frac{1}{2b}\sqrt{\nabla}\,[d\,(d-a) + 2bc) + d\sqrt{\nabla}\,],$$

and the equation for K becomes

$$\frac{d\,(d-a)+2bc-d\sqrt{\nabla}}{ad-bc}=\frac{-4bc}{d\,(d-a)+2bc+d\sqrt{\nabla}}\,;$$

that is,

$$[d\,(d-a)+2bc]^2-d^2\,[(d-a)^2+4bc]=-4bc\,(ad-bc),$$

which is right; and similarly the equation for L leads to this same equation.

11. We may now establish, on the principles appearing in No. 5, the different forms of the system. Using ideas and aid as abbreviations for idempotent and nilpotent respectively, there are in all 11 cases.

(i) 3 idems. Taking two of these to be x and y, the system is

$$x^2=x,\quad xy=ex+dy,\quad yx=ex+fy,\quad y^2=y.$$

Hence $\Omega=(1-c-e)\,x^2y+(1-d-f)\,xy^2=0$, so that the third factor is

$$(1-c-e)\,x+(1-d-f)\,y.$$

This must not reduce itself to x or y, for, if so, there would be a twofold idem.; viz. as negative conditions we must have $c+e\neq 1$, $d+f\neq 1$.

And we have

$$[(1-c-e)\,x+(1-d-f)\,y]^2=[1-(c+e)\,(d+f)]\,[(1-c-e)\,x+(1-d-f)\,y],$$

which must be an idem.; viz. we have the further negative condition $(c+e)\,(d+f)\neq 1$.

(ii) 2 idems and 1 nil. This arises from (i) by assuming therein

$$(c+e)\,(d+f)=1,\ \text{say}\ d+f=\frac{1}{c+e}\,;$$

for then, writing $z=-\,(c+e)\,x+y$, we have

$$\begin{aligned}z^2&=(c+e)^2\,x^2-(c+e)\,[(c+e)\,x+(d+f)\,y]+y,\\&=[1-(c+e)\,(d+f)]\,y,\ =0\,;\end{aligned}$$

viz. z is a nil. And, if in the equations instead of the idem y we introduce the nil z, then the equations assume the form

$$x^2=x,\quad xz=[(c+e)\,d-c]\,x+dz,\quad zx=[(c+e)\,f-c]\,x+fz,\quad z^2=0\,;$$

with the idem y, $=(c+e)\,x+z$, we have the negative conditions $c+e\neq 0$, implying $d+f\neq 0$.

But the equations are obtained in a more simple form by taking a for the idem and y for the nil; viz. we then have $x^2=x$, $xy=ax+dy$, $yx=ex+fy$, $y^2=0$: we must then have z, $=-(c+e)\,x+(1-d-f)\,y$, for an idem; this gives $z^2=-(c+e)\,(d+f)\,z$, and we have the negative conditions $c+e\neq 0$, $d+f\neq 0$ or 1.

(iii) 1 idem and 2 nils. This may be deduced from (ii) by writing therein $d+f=0$; for then $z, = -(c+e)\,x+y$, is a nil. The equations are $x^2=x$, $xy=ax-dy$, $y^2=0$: and if, instead of the idem y, we introduce therein z by the equation $z=-(c+e)\,x+y$, the equations become $x^2=x$, $yz=[-(c+e)\,d+a]\,x+dz$, $zx=[-(c+e)\,f+c]\,x+cz$, $z^2=0$, with the negative condition $c+e\neq 0$.

But it is more simple to take x, y as the nils: the equations then are $x^2=0$, $xy=cx+dy$, $yx=ex+fy$, $y^2=0$. We must have $z, = ex+cy$, for an idem; this gives $z^2=(c+e+f+d)\,xy$, becomes $=z$ and $=y$; so that x or y is a twofold nil. Or, what comes to the same thing, we have

$$\Omega = -(c+e)\,x^2y-(d+f)\,xy^2,$$

and Ω has as twofold factor if $c+e=0$ or $d+f=0$.

(iv) A twofold idem and a onefold idem. Taking x for the twofold idem and y for the onefold idem, Ω must reduce itself to $(1-c-e)\,x^2y$, we must have $d+f=1$, or say $f=1-d$. The equations are $x^2=x$, $xy=cx+dy$, $yx=ex+(1-d)\,y$, $y^2=y$; and we have the negative conditions $c+e\neq 0$, for otherwise Ω would vanish identically.

(v) A twofold idem and a onefold nil. Taking x for the twofold idem and y for the onefold nil, the equations are $x^2=x$, $xy=cx+dy$, $yx=ex+fy$, $y^2=0$; and we have the negative conditions $c+e\neq 0$, $d+f\neq 0$.

(vi) A twofold nil and a onefold idem. Taking those to be x and y, then $d+f=0$, and the equations are $x^2=0$, $xy=cx+dy$, $yx=ex-dy$, $y^2=y$; we have the negative condition $c+e\neq 0$.

(vii) A twofold nil and a onefold nil. Taking those to be x and y, we have $d+f=0$ and the equations are $x^2=0$, $xy=cx+dy$, $yx=ex-dy$, $y^2=0$; with the negative condition $c+e\neq 0$.

(viii) A threefold idem. Taking this to be x, then Ω must reduce itself to gx^3, viz. we must have $h=c+e$, $1=d+f$; and the equations are $x^2=x$, $xy=cx+dy$, $yx=ex+(c+e)\,y$, $y^2=gx+(c+e)\,y$; we have the negative condition $g\neq 0$, for otherwise Ω would vanish identically.

(ix) A threefold nil. Taking this to be x, then we must have $h=c+e$, $0=d+f$; the equations are $x^2=0$, $xy=cx+dy$, $yx=ex-dy$, $y^2=gx+(c+e)\,y$; and there is again the negative condition $g\neq 0$.

(x) $\Omega=0$ identically; infinity of idems, 1 nil. Ω will vanish identically if $g=0$, $h=c+e$, $a=d+f$, $b=0$. If there is 1 idem, there will be no infinity of idems, and 1 nil. For, assume an idem x, $x^2=x$; and, if possible, let there be no other idem; then there will be a nil y, $y^2=0$. We have $c+e=0$, $d+f=1$; and the equations

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are $x^2 = x$, $xy = cx + dy$, $yx = -cx + (1-d)\,y$, $y^2 = 0$; whence $xy + yx = y$. Taking α, β arbitrary coefficients, we have

$$(\alpha x + \beta y)^2 = \alpha^2 x + \alpha\beta y, \ = \alpha\,(\alpha x + \beta y)\,;$$

hence for $\alpha x + \beta y$ to be an idem, except in the case $\alpha = 0$, when it is the original nil y.

If besides the idem x we have an idem y, then the conditions are $c+e = 1$, $d + f = 1$; the equations are

$$x^2 = x, \quad xy = cx + dy, \quad yx = (1-c)\,x + (1-d)\,y, \quad y^2 = y\,;$$

whence $xy + yx = x + y$. Considering the combination $\alpha x + \beta y$, we have

$$(\alpha x + \beta y)^2 = \alpha^2 x + \alpha\beta\,(x + y) + \beta^2 y, \ = (\alpha + \beta)\,(\alpha x + \beta y).$$

This is an idem, except in the case $\alpha + \beta = 0$, when it is a nil; or, say we have the single nil $x - y$. We have thus again an infinity of idems, 1 nil.

(xi) $\Omega = 0$ identically; an infinity of nils. Taking the two nils x and y, the conditions are $c + e = 0$, $d + f = 0$, the equations are $x^2 = 0$, $xy = cx + dy$, $yx = -cx - dy$, $y^2 = 0$; whence $xy + yx = 0$, or there are an infinity of nils.

viz. $\alpha x + \beta y$ is a nil; or there are an infinity of nils.

12. The different cases may be grouped together as follows:—

A. 2 idems, (i), (ii), (iv), (x).

Equations $x^2 = x,\ xy = cx + dy,\ yx = ex + fy,\ y^2 = y.$

B. 1 idem and 1 nil, (ii), (iii), (v), (vi), (x).

Equations $x^2 = x,\ xy = cx + dy,\ yx = ex + fy,\ y^2 = 0.$

C. 2 nils, (iii), (vii), (xi).

Equations $x^2 = 0,\ xy = cx + dy,\ yx = ex + fy,\ y^2 = 0.$

D. Threefold idem, (viii).

Equations $x^2 = x,\ xy = cx + dy,\ yx = ex - (1-d)\,y,\ y^2 = gx + (c+e)\,y.$

E. Threefold nil, (ix).

Equations $x^2 = 0,\ xy = cx + dy,\ yx = ex - dy,\ y^2 = gx + (c+e)\,y.$

The several cases of A, B, C respectively are distinguished by negative conditions which need not be here repeated.

13. I consider, as in my Note before referred to, the conditions in order that the system may be associative. We have the 8 products, x^3, x^2y, xyx, xy^2, yx^2, yxy, y^2x, y^3, giving rise to equations $x\,.\,x^2 = x^2\,.\,x$, $x\,.\,xy = xy\,.\,x$, $\ldots$, $y\,.\,y^2 = y^2\,.\,y$, which, on putting therein for x^2, xy, yx, y^2 their values, must be satisfied identically. We thus obtain in

the first instance 16 relations, but some of these are repeated, and we have actually only 12 relations; viz. the relations are

(twice)	$b\,(c-e)=0,$
	$b\,(f-d)=0,$
	$g\,(c-e)=0,$
(twice)	$g\,(f-d)=0,$
(twice)	$bg-cd=0,$
(twice)	$bg-ef=0,$
	$c\,(e-h)+g\,(d-a)=0,$
	$d\,(d-a)+b\,(c-b)=0,$
	$e\,(e-h)+g\,(f-a)=0,$
	$b\,(e-h)+f\,(f-a)=0,$
	$a\,(e-c)-cf+de=0,$
	$h\,(f-d)-cf+de=0.$

14. From the first four equations it appears that either $b=0$ and $g=0$, or else $c=e$ and $d=f$. I attend first to the latter case, viz. we here have the commutative system

$$x^2=ax+by,\quad xy=yx=cx+dy,\quad y^2=gx+hy.$$

In order that this may be associative, we must still have the relations

$$\begin{aligned} bg-cd&=0,\\ c\,(c-h)+g\,(d-a)&=0,\\ d\,(d-a)+b\,(c-b)&=0,\end{aligned}$$

or, as they may also be written,

$$\left|\begin{array}{ccc} b, & -c, & d-a\\ -d, & g, & c-b\end{array}\right|=0.$$

These are satisfied by $g=\dfrac{cd}{b}$, $h=\dfrac{d^2+bc-cd}{b}$, and we have thus the commutative and associative system of the Note.

Every system is of the form A, B, C, D, or E; and it can be shown that the commutative and associative system is not of the form D. For, if D were commutative, we should have $c=e$, $d=\frac{1}{2}$, viz. the equations will be $x^2=x$, $xy=yx=cx+\frac{1}{2}y$, $y^2=gx+2cy$, that is,

$$a,\ b,\ c,\ d,\ y,\ h=1,\ 0,\ c,\ \tfrac{1}{2},\ 2c;$$

and the last of the three relations, viz. $d\,(d-a)+b\,(c-b)=0$, would thus be $\frac{1}{2}\,(\frac{1}{2}-1)=0$, which is not satisfied. Hence the commutative and associative system can only be of one of the forms A, B, C, and E.

15. First, of the form be A, B, or C ; there will be the two idem-or-nil symbols x and y, that is, we may assume $b = 0$, $g = 0$, and the associative conditions then become $cd = 0$, $c\,(c - h) = 0$, $d\,(d - a) = 0$, viz. for the forms A, B, C,

$$\begin{aligned} &\text{A.} \quad x^2 = x, \quad y^2 = y, \\ &\text{B.} \quad x^2 = x, \quad y^2 = 0, \\ &\text{C.} \quad x^2 = 0, \quad y^2 = 0. \end{aligned}$$

these are $cd = 0$, $c\,(c - 1) = 0$, $d\,(d - 1) = 0\,;$ $c = 0$ or 1, $d = 0$ or 1,

$$\begin{aligned} &,, \quad cd = 0, \quad c^2 = 0, \quad d\,(d-1) = 0\,; \; c = 0, \quad d = 0 \text{ or } 1, \\ &,, \quad cd = 0, \quad c^2 = 0, \quad d^2 = 0\,; \quad c = 0, \quad d = 0. \end{aligned}$$

But for the form A, if $c = 0$, $d = 1$, then is $xy = yx = y$, then, writing $z = x - y$, we have $z^2 = x$, $yz = zy = 0$, then, writing $x' = -y$, $z = z$, we have $x'^2 = -x'$, then $x' = -x$, that is, in each of these is reduced to the first case $c = 0$, $d = 0$, that is, $x^2 = x$, $xy = 0$, $y^2 = y$.

For the form B, if $c = 0$, $d = 1$, the system is $x^2 = x$, $xy = yx = y$, $y^2 = 0$; and this cannot be reduced to the first case $x^2 = x$, $xy = yx = 0$, $y^2 = 0$.

For the form C, the system is $x^2 = 0$, $xy = yx = 0$, $y^2 = 0$. In each of these is only the commutative, but not associative, conditions $b = 0$, $g = 0$, as above.

For the form E, here we have $a = 0$, $b = 0$, $c = e$, $d = f$, and the system may be commutative, $h = 2c$, viz. the equations must be $x^2 = 0$, $xy = yx = cx$, $y^2 = gx + 2cy$. The associative conditions then give $c = 0$; the equations are $x^2 = 0$, $xy = yx = 0$, $y^2 = gx$. Writing $\dfrac{x}{g}$ instead of x, and for convenience interchangingly x and y, the equations are

$$x^2 = y, \quad xy = yx = 0, \quad y^2 = 0.$$

16. The commutative associative system is thus seen to be reducible as follows:—

A system is $x^2 = x,\ xy = yx = 0,\ y^2 = y,$ first mixed system,*see* No. 2.

B. ,, $x^2 = x,\ xy = yx = y,\ y^2 = 0,$ Peirce's system (a_1).

or else

B. ,, $x^2 = x,\ xy = yx = 0,\ y^2 = 0,$ second mixed system.

C. ,, $x^2 = 0,\ xy = yx = 0,\ y^2 = 0,$ third mixed system.

E. ,, $x^2 = y,\ xy = yx = 0,\ y^2 = 0,$ Peirce's system (c_1).

I said, at the end of my Note before referred to, that it had been pointed out to me "that my system [the commutative associative system], in the general case $ad - bc$ not $= 0$, is expressible as a mixture of two algebras of the form (a_1), see *American Journal of Mathematics*, vol. iv., p. 120; whereas, if $ad - bc = 0$, it is reducible to the form (a), see p. 122 (l.c.)." The accurate conclusion is as above, that the commutative associative system is either a mixed system of one of the three forms, or else a system (a_1), or (c_1).

17. Considering next the non-commutative associative system, we have here, *vide* No. 14, $b = 0$, $g = 0$; and the relations which remain to be satisfied then are

$$cd = 0,\ ef = 0,\ c\,(c - h) = 0,\ e\,(e - h) = 0,\ d\,(d - a) = 0,\ f\,(f - a) = 0,$$

$$a\,(e - c) - cf + de = 0,\ h\,(f - d) - cf + de = 0.$$

The first equation gives $cd = 0$, viz. $c = 0$ or $d = 0$; but we may attend exclusively to the case $c = 0$, for the case $d = 0$ may be deduced from this by the interchange of x, y. We have then $ef = 0$; and it will be convenient to separate the cases

I. $c = 0,\ e \neq 0,\ x^2 = ax,\ xy = yx = 0,\ y^2 = hy$;

II. $c = 0,\ e \neq 0,\ f = 0,$;

III. $c = 0,\ e = 0,\ f \neq 0,$.

I. (a) $d = 0$; $x^2 = ax,\ xy = yx = 0,\ y^2 = hy$: commutative, and so included in what precedes.

I. (b) $d = a,\ h = 0$; $x^2 = ax,\ xy = ay,\ yx = 0,\ y^2 = 0$, which is Peirce's system (b_1).

I. (c) $d = a,\ h = a$; $x^2 = ax,\ xy = ay,\ yx = 0,\ y^2 = ay$: commutative, and so included in what pr

II. (d) $d = 0,\ f = a$: $x^2 = ax,\ xy = 0,\ yx = ay,\ y^2 = 0$, writing as we may do $a = 1$, this is $x^2 = x$

II. (e) $d = a,\ e = h$: $x^2 = ax,\ xy = ay,\ yx = hx,\ y^2 = hy$, writing as we may do $a = 1,\ h = 1$, this

Thus x, z form the system $x^2 = x$, $xz = z$, $zx = 0$, $z^2 = 0$ (or, what is the same thing, y, z form a system $y^2 = y$, $yz = 0$, $zy = z$, $z^2 = 0$); each of these is Peirce's system (b_1).

The conclusion is that every non-commutative associative system is either Peirce's system (b_1), or else the omitted system (d_1). Hence, disregarding the mixed systems, every associative system is either (a_1), (b_1), (c_1), or (d_1).

19. It may be proper to show that the systems (b_1), $x^2 = x$, $xy = y$, $yx = 0$, $y^2 = 0$, and (d_1), $x^2 = x$, $xy = 0$, $yx = y$, $y^2 = 0$, or say

	a	b	c	d	e	f	g	h
(b_1)	1	0	0	1	0	0	0	0
(d_1)	1	0	0	0	0	1	0	0

are really distinct from each other. Observe that they each belong to the case (x), $\Omega = 0$, an infinity of idoms and 1 nil; viz. in each of them writing $z = x + \beta y$, β an arbitrary coefficient, we have $z^2 = x^2 + \beta\,(xy + yx) + \beta^2 y^2$, $= x + \beta y$, $= z$, we have z an idom, and y is the only nil. And, this being so, in the second system, $xy = 0$, $yx = y$; viz. the system is $x^2 = x$, $xy = 0$, $yx = y$, $y^2 = 0$, retaining, when we write therein z for x, its original form. And similarly, in the second system, $xy = 0$, $yx = y$; viz. the system is $x^2 = x$, $xy = y$, $yx = 0$, $y^2 = 0$, retaining, when we write therein z for x, its original form. The two are thus distinct systems, in no wise transformable the one into the other.

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THE BINOMIAL EQUATION $x^p - 1 = 0$; QUINQUISECTION. SECOND PART.

[From the *Proceedings of the London Mathematical Society*, vol. xvi. (1885), pp. 61–63.]

In the paper, "The binomial equation $x^p - 1 = 0$; quinquisection," *Proc. Lond. Math. Soc.*, t. xii. (1881), pp. 15, 16, [744], I considered for an exponent $p = 5n + 1$, the five periods X, Y, Z, W, T connected by the equations

$$\begin{array}{rcccccc} & & X, & Y, & Z, & W, & T \\ X^2 & = & a, & b, & c, & d, & e \\ XY & = & f, & g, & h, & i, & j \\ XZ & = & k, & l, & m, & n, & o, \end{array}$$

and the equations deduced from these by cyclical permutations of the periods and of the coefficients of each set; but I did not obtain completely even the linear relations connecting the coefficients. I since found, by induction from the examples given in the Table 1, that the coefficients could be expressed linearly in terms of the linearly independent integer numbers a, β, f, k as follows: viz. introducing for convenience the new number θ, such that

$$a + \beta + \theta = \tfrac{1}{5}(p - 1),$$

then the expressions in question are

$$\begin{array}{rcccccc} a,\, b,\, c,\, d,\, e & = & -1 - 2\theta + a + \beta, & -\theta - a + \beta + f, & -\theta - a + \beta + k, & -a - 2\beta - k, & -2a - \beta - \\ f,\, g,\, h,\, i,\, j & = & f, & \theta - a - f, & a, & \beta, & a, \\ k,\, l,\, m,\, n,\, o & = & k, & a, & \theta - \beta - k, & \beta, & a, \end{array}$$

and I found further that, substituting these values of the coefficients in the 20 quadric relations referred to in the former paper, the 20 relations reduced themselves to two equations only, viz. these were

$$\theta\,(-2\alpha + \beta + k) + 3a^2 - \beta^2 + \alpha\,(f - k - 1) - \beta f + f^2 - 2fk = 0,$$
$$-\theta^2 + \theta\,(3\beta + 2k + f) + a^2 - a\beta - 3\beta - \alpha k + \beta\,(1 - f - k) - k^2 - 2fk = 0.$$

The final result thus is that the coefficients are expressed as functions of the five numbers a, β, f, k, θ, connected by the linear equation $\alpha+\beta+\theta = \frac{1}{5}(p-1)$, and the two quadric equations. I remark that formulae equivalent to these were obtained and proved by Mr F. S. Carey in his Trinity Fellowship Dissertation, 1884; viz. writing $n=\frac{1}{5}(p-1)$, his formulae were

$$\begin{array}{rcc} a,\ b,\ c,\ d,\ e & = & n-a,\ \beta-n,\ \gamma-n,\ \delta-n,\ \varepsilon-n, \\ f,\ g,\ h,\ i,\ j & = & f,\ \theta-a-f,\ \alpha,\ \beta,\ \rho, \\ k,\ l,\ m,\ n,\ o & = & \gamma,\ \alpha,\ \delta,\ \beta,\ \sigma,\ \alpha, \end{array}$$

with the three linear relations

$$\begin{gathered} \alpha+\beta+\gamma+\delta+\varepsilon = n-1, \\ \beta+\varepsilon+2\rho+\sigma = n, \\ \gamma+\delta+\rho+2\sigma = n, \end{gathered}$$

and the two quadric relations

$$\begin{gathered} \delta^2+\gamma^2+2\sigma a+(\rho-\sigma)(\delta+\gamma)-2\rho(\rho+\sigma) = (\delta-\gamma)(\beta-\varepsilon), \\ \beta^2+\varepsilon^2+2\rho a+(\sigma-\rho)(\beta+\varepsilon)-2\sigma(\rho+\sigma) = (\gamma-\delta)(\beta-\varepsilon), \end{gathered}$$

the coefficients being thus expressed in terms of the seven numbers a, β, γ, δ, ε, ρ, σ connected by five equations. The equivalence of the two sets of formulae may be shown without difficulty.

To the Table 2 of the Quintic Equations, given in the paper, may be added the following result from Legendre's *Théorie des Nombres*, Ed. 3, t. ii., p. 213,

$$\begin{array}{c|ccccccc} p & q^6 & q^5 & q^4 & q^3 & q^2 & q & 1 \\ \hline 641 & 1 & +1 & -256 & -564 & +3258 & -3120 & \end{array} = 0,$$

calculated by him for the isolated case $p=641$.

$$\frac{x}{1-\beta y}+\frac{y}{1-\gamma z}+\frac{z}{1-\alpha\beta}=0 \quad \text{and} \quad \frac{\xi}{a\,(\gamma-\beta)}+\frac{\eta}{\beta\,(\gamma-a)}+\frac{\zeta}{\gamma\,(\beta-a)}=0$$

respectively, we have 6+6 equations of like forms; but these two systems each of 6 equations are equivalent to each other, so that instead of $6+16+3+3=28$, we have $6+16+6\,(=6)=28$ equations for the 28 bitangents[1]. I make another slight change of notation by introducing the single letters f, g, h to denote the reciprocals $\frac{1}{a}$, $\frac{1}{b}$, $\frac{1}{c}$; and I consider the whole question as follows. The theory is based on the equations

$$\begin{aligned}
ax+by+cz+f\xi+g\eta+h\zeta&=0,\\
a_1x+b_1y+c_1z+f_1\xi+g_1\eta+h_1\zeta&=0,\\
a_2x+b_2y+c_2z+f_2\xi+g_2\eta+h_2\zeta&=0,\\
a_3x+b_3y+c_3z+f_3\xi+g_3\eta+h_3\zeta&=0,
\end{aligned}$$

where $af=bg=ch=a_1f_1$ &c., $=a_3h_3$, $=1$: and the coefficients a, b, c, a_1, b_1, c_1, a_2, b_2, c_2 are arbitrary. The equations $x=0$, $y=0$, $z=0$ represent any three given lines: and

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ON THE BITANGENTS OF A PLANE QUARTIC.

[From *Crelle's Journal der Mathem.*, t. xciv. (1883), pp. 93–115; *Camb. Phil. Soc. Proc.*, t. iv. (188

Riemann in the paper "Zur Theorie der Abelschen Functionen für den Fall $p=3$," *Werke*, pp. 456–479, has given a remarkably elegant solution of the problem of the bitangents of a plane quartic. But his formulae may be improved by a slight change; viz. we may in his first equation $x+y+z+f\xi+g\eta+h\zeta=0$ introduce coefficients so as to bring this into the same form as the other three equations. It thus appears that, instead of his $3+3$ equations of the forms

$$\frac{x}{1-\beta y}+\frac{y}{1-\gamma z}+\frac{z}{1-\alpha\beta}=0 \quad \text{and} \quad \frac{\xi}{a\,(\gamma-\beta)}+\frac{\eta}{\beta\,(\gamma-a)}+\frac{\zeta}{\gamma\,(\beta-a)}=0$$

[1] It will appear further on that the equation of each of the last-mentioned 6 bitangents can be expressed in 6 different forms.

[1] It will appear further on that the equation of each of the last-mentioned 6 bitangents can be expressed in 6 different forms.

considering ξ, η, ζ to be determined as linear functions of x, y, z by means of the first three equations, then (the coefficients a, b, c, a_1, b_1, c_1, a_2, b_2, c_2 being determined accordingly) the equations $\xi = 0$, $\eta = 0$, $\zeta = 0$ will also represent any three given lines. But observe that, if the equation of the first of these lines is $\alpha x + my + nz = 0$, ξ is not $=$ an arbitrary multiple $\theta\,(\alpha + my + nz)$ of the linear function $\alpha + my + nz$, but is $= G\,(\alpha + my + nz)$, G being a completely determinate value: and the like as regards η and ζ respectively.

The coefficients a_3, b_3, c_3 of the fourth equation are such that this fourth equation is a mere consequence of the other three: viz. we must have

$$\begin{aligned} a_3 &= \lambda a + \lambda_1 a_1 + \lambda_2 a_2, \\ b_3 &= \lambda b + \lambda_1 b_1 + \lambda_2 b_2, \\ c_3 &= \lambda c + \lambda_1 c_1 + \lambda_2 c_2, \\ f_3 &= \lambda f + \lambda_1 f_1 + \lambda_2 f_2, \\ g_3 &= \lambda g + \lambda_1 g_1 + \lambda_2 g_2, \\ h_3 &= \lambda h + \lambda_1 h_1 + \lambda_2 h_2, \end{aligned}$$

or, what is the same thing, we must have

$$\left\|\begin{array}{cccccc} a, & b, & c, & f, & g, & h \\ a_1, & b_1, & c_1, & f_1, & g_1, & h_1 \\ a_2, & b_2, & c_2, & f_2, & g_2, & h_2 \\ a_3, & b_3, & c_3, & f_3, & g_3, & h_3 \end{array}\right\| = 0,$$

viz. each of the determinants formed with four columns out of this matrix is equal to 0.

Using the equations in λ, λ_1, λ_2, and writing for shortness

$$\begin{aligned} a_3 f_3 = a_0 f_0, &\quad a_3 g_3 + a_0 f_1 = a f_1 + a_1 f_0 = F,\ F_1,\ F_2, \\ b_3 g_3 = b_0 g_0, &\quad b_3 g_3 + b_0 g_1 = b g_1 + b_1 g_0 = G,\ G_1,\ G_2, \\ c_3 h_3 = c_0 h_0, &\quad c_3 h_3 + c_0 h_1 = c h_1 + c_1 h_0 = H,\ H_1,\ H_2\,; \end{aligned}$$

then, forming the products $a_3 f_3$, $b_3 g_3$, $c_3 h_3$, we find

$$\begin{aligned} 1 - \lambda^2 - \lambda_1^2 - \lambda_2^2 &= P\lambda\lambda_1 + F_1\lambda\lambda_2 + F_2\lambda_1\lambda_2, \\ ,, &= G\lambda\lambda_1 + G_1\lambda\lambda_2 + G_2\lambda_1\lambda_2, \\ ,, &= H\lambda\lambda_1 + H_1\lambda\lambda_2 + H_2\lambda_1\lambda_2, \end{aligned}$$

or, as these equations may be written,

$$\lambda\lambda_1 : \lambda\lambda_2 : \lambda_1\lambda_2 = \left|\begin{array}{ccc} 1, & F, & F_2 \\ 1, & G, & G_2 \\ 1, & H, & H_2 \end{array}\right| : \left|\begin{array}{ccc} 1, & F, & F_1 \\ 1, & G, & G_1 \\ 1, & H, & H_1 \end{array}\right| : \left|\begin{array}{ccc} 1, & F_1, & F_2 \\ 1, & G_1, & G_2 \\ 1, & H_1, & H_2 \end{array}\right|.$$

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say for shortness

$$= \Pi : \Lambda : \Lambda_1 : \Lambda_2 .$$

We have thus

$$\frac{\lambda^2}{1-\lambda^2-\lambda_1^2-\lambda_2^2} = \frac{\Lambda_1\Lambda_2}{\Pi\Lambda},$$

$$\frac{\lambda_1^2}{1-\lambda^2-\lambda_1^2-\lambda_2^2} = \frac{\Lambda\Lambda_2}{\Pi\Lambda_1},$$

$$\frac{\lambda_2^2}{1-\lambda^2-\lambda_1^2-\lambda_2^2} = \frac{\Lambda\Lambda_1}{\Pi\Lambda_2},$$

and thence

$$\frac{1}{1-\lambda^2-\lambda_1^2-\lambda_2^2} = \frac{1}{\Pi}\left(\frac{\Lambda_1\Lambda_2}{\Lambda} + \frac{\Lambda\Lambda_2}{\Lambda_1} + \frac{\Lambda\Lambda_1}{\Lambda_2}\right),$$

equations which give λ, λ_1, λ_2, and consequently a_3, b_3, c_3, f_3, g_3, h_3 in terms of known quantities, the several expressions depending on the single radical

$$\sqrt{\frac{1}{\Pi}\left(\frac{\Lambda_1\Lambda_2}{\Lambda} + \frac{\Lambda\Lambda_2}{\Lambda_1} + \frac{\Lambda\Lambda_1}{\Lambda_2}\right)}.$$

Observe that we have rationally

$$\lambda : \lambda_1 : \lambda_2 = \frac{1}{\Lambda} : \frac{1}{\Lambda_1} : \frac{1}{\Lambda_2}.$$

I consider now the quartic curve

$$\sqrt{x\xi} + \sqrt{y\eta} + \sqrt{z\zeta} = 0,$$

and I write down the equations of the 28 bitangents, each with its triple-theta characteristic as given by Riemann, and also with its dual symbol, derived from Hesse's method. I assume that these characteristics and dual symbols are properly attached to the several lines: and I insert also the current nos. 1 to 28. The equations are:

Current No.	Dual Symbol	Charac-teristic	
1	18	111 111	$x = 0,$
2	28	001 011	$y = 0,$
3	38	011 001	$z = 0,$
4	23	010 010	$\xi = 0,$
5	15	100 110	$\eta = 0,$
6	12	110 100	$\zeta = 0,$

Current No.	Dual Symbol	Charac-teristic	
7	49	101 100	$ax + by + cz = 0,\ f\xi + g\eta + h\zeta = 0,$
8	14	010 011	$f_1\xi + b_1y + c_1z = 0,\ a_1x + g_1\eta + h_1\zeta = 0,$
9	24	100 011	$a_2x + g_2\eta + c_2z = 0,\ f_2\xi + b_2y + h_2\zeta = 0,$
10	34	100 101	$a_3x + b_3y + h_3\zeta = 0,\ f_3\xi + g_3\eta + c_3z = 0,$
11	58	100 010	$a_1x + f_1\xi + c_1z = 0,$
12	16	001 010	$f_1\xi + b_1y + h_1\zeta = 0,$
13	25	101 010	$a_2x + g_2\eta + h_2\zeta = 0,$
14	35	111 110	$a_3x + g_3\eta + h_3\zeta = 0,$
15	68	110 010	$a_1x + b_1y + h_1\zeta = 0,$
16	16	001 101	$f_1\xi + g_1\eta + h_1\zeta = 0,$
17	26	111 101	$a_2x + b_2y + h_2\zeta = 0,$
18	36	101 011	$a_3x + b_3y + h_3\zeta = 0,$
19	78	010 110	$a_1x + b_1y + c_1z = 0,$
20	17	101 110	$f_1\xi + b_1y + c_1z = 0,$
21	27	011 110	$a_2x + f_2\xi + c_2z = 0,$
22	37	000 111	$a_3x + b_3y + h_3\zeta = 0,$
23	67	100 100	$\frac{x}{bc-b_1c_1} + \frac{y}{ca-c_1a_1} + \frac{z}{ab-a_1b_1} = 0,\ \frac{1}{c_2h_2-c_3h_3} + \frac{1}{b_2f_2-b_3f_3} + \frac{1}{f_2g_2-f_3g_3} = 0,$ $\frac{x}{bc-b_2c_2} + \frac{y}{ca-c_2a_2} + \frac{z}{ab-a_2b_2} = 0,\ \frac{1}{c_1h_1-c_3h_3} + \frac{1}{a_1h_1-a_3h_3} + \frac{1}{f_1g_1-f_3g_3} = 0,$
24	57	110 001	$\frac{x}{bc-b_3c_3} + \frac{y}{ca-c_3a_3} + \frac{z}{ab-a_3b_3} = 0,$
25	56	010 001	$\frac{x}{bc-b_1c_1} + \frac{y}{ca-c_1a_1} + \frac{z}{ab-a_1b_1} = 0,$
26	45	000 011	$\frac{x}{b_2f_2-b_3f_3} = \frac{y}{c_2g_2-c_3g_3} = \frac{z}{a_2h_2-a_3h_3},$
27	46	001 110	$\frac{x}{b_1f_1-b_3f_3} = \frac{y}{c_1g_1-c_3g_3} = \frac{z}{a_1h_1-a_3h_3},$
28	47	111 010	$\frac{x}{b_1f_1-b_2f_2} = \frac{y}{c_1g_1-c_2g_2} = \frac{z}{a_1h_1-a_2h_2},$

As regards each of the bitangents 7 to 22, the equivalence of the two forms of equation is at once seen by means of the fundamental equations $ax + by + cz + f\xi + g\eta + h\zeta = 0$: as regards the remaining bitangents 23 to 28, the equation of each of these is expressible in eight different forms, which are written down in full for the bitangent 23; the equivalence of the eight forms of equation must of course be proved.

I proceed to show that each of the 28 lines is, in fact, a bitangent to the quartic curve

$$\sqrt{x\xi} + \sqrt{y\eta} + \sqrt{z\zeta} = 0,$$

or, what is the same thing, that writing this equation in the form

$$\Omega = x^2\xi^2 + y^2\eta^2 + z^2\zeta^2 - 2yz\eta\zeta - 2zx\zeta\xi - 2xy\xi\eta = 0,$$

then that by means of any one of these equations Ω becomes a perfect square.

This is obviously the case for each of the equations 1, 2, 3, 4, 5, 6.

For the equations 7, 8, 9, 10, observe that the equation of the quartic may be written:

$$\sqrt{axf\xi} + \sqrt{byg\eta} + \sqrt{czh\zeta} = 0,$$

or, what is the same thing,

$$\Omega = ax^2\xi^2, f^2\xi^2 + \ldots = 0.$$

Hence writing x, y, z, ξ, η, ζ in place of ax, by, cz, $f\xi$, $g\eta$, $h\zeta$, so that now

$$x + y + z + \xi + \eta + \zeta = 0,$$

the equation of the line 7 is expressed in the two equivalent forms $x+y+z = 0$, $\xi + \eta + \zeta = 0$. We have for Ω its original value

$$\Omega = x^2\xi^2 + y^2\eta^2 + z^2\zeta^2 - 2yz\eta\zeta - 2zx\zeta\xi - 2xy\xi\eta,$$

$= (x\xi - y\eta - z\zeta)^2$, a perfect square. The like proof applies to the equations 8, 9, 10; and then writing herein $x = -x - y$, $\xi = -\xi - y\eta$, we have $x\xi - y\eta - z\zeta = xy + y\xi$ and consequently $\Omega = (x\eta + y\xi)^2 = (\xi y - y\xi)^2$, a perfect square. The like proof applies to the equations 8, 9, 10; and then writing herein $z = -x - y$, $\zeta = -\xi - \eta$, we have to apply this result to the equations 11, 12, 13, 14; 15, 16, 17, 18; 19, 20, 21, 22.

We come now to the equation of the line 23, say this is in the first instance taken to be

$$\frac{x}{bc - b_1c_1} + \frac{y}{ca - c_1a_1} + \frac{z}{ab - a_1b_1} = 0\,;$$

if from this equation, by means of the two fundamental equations

$$ax + by + cz + f\xi + g\eta + h\zeta = 0,$$
$$a_1x + b_1y + c_1z + f_1\xi + g_1\eta + h_1\zeta = 0,$$

we eliminate first (y,z), secondly (z,x), and thirdly (x,y), it is found that ξ, η, ζ will also disappear in the three cases respectively, and that the resulting equations are

$$\begin{aligned}
\frac{x}{bc-b_1c_1}+\frac{\eta}{bb_1\,(ca-c_1a_1)}+\frac{\zeta}{cc_1\,(ab-a_1b_1)}&=0,\\
-\frac{y}{aa_1\,(bc-b_1c_1)}+\frac{\eta}{ca-c_1a_1}+\frac{\zeta}{cc_1\,(ab-a_1b_1)}&=0,\\
\frac{\xi}{aa_1\,(bc-b_1c_1)}-\frac{\eta}{bb_1\,(ca-c_1a_1)}+\frac{z}{ab-a_1b_1}&=0,
\end{aligned}$$

so that each of these is, in fact, equivalent to the original equation in (x,y,z): and observe that, adding together the three equations, we reproduce the original equation in (x,y,z).

Write for shortness

$$\begin{aligned}
P&=bc-b_1c_1, &\mathfrak{a}&=bc-b_1c_1,\\
Q&=ca-c_1a_1, &\mathfrak{B}&=ca-c_1a_1,\\
R&=ab-a_1b_1, &\gamma&=ab-a_1b_1,
\end{aligned}$$

then the equation in (x,y,z) is

$$QRx+RPy+PQz=0,$$

so that, eliminating y and z, we find

$$\begin{vmatrix} b, & c, & ax+f\xi+g\eta+h\zeta\\ b_1, & c_1, & a_1x+f_1\xi+g_1\eta+h_1\zeta\\ RP, & PQ, & QRx \end{vmatrix}=0.$$

In this equation, the coefficient of x is

$$\begin{aligned}
&(bc_1-b_1c)\,QR+(cm_1-c_1a)\,RP+(ab_1-a_1b)\,PQ,\\
&\quad=(bc_1-b_1c)\,[aPb+a_1b_1c_1-aa_1\,(bc_1-b_1c)]\\
&\quad\;\;+(cm_1-c_1a)\,[bP+a_1b_1c_1-bb_1\,(ca_1-c_1a)]\\
&\quad\;\;+(ab_1-a_1b)\,[cP+a_1b_1c_1-cc_1\,(ab_1-a_1b)],\\
&=-aa_1\,(bc_1-b_1c)^2-bb_1\,(ca_1-c_1a)^2-cc_1\,(ab_1-a_1b)^2,\\
&\qquad=-aa_1\gamma.
\end{aligned}$$

The coefficient of ξ is

$$=RP\,(cf_1-c_1f)-PQ\,(bf_1-b_1f),$$

which (observing that $cf_1-c_1f=\frac{c}{a_1}-\frac{c_1}{a}=\frac{1}{aa_1}Q$ and $bf_1-b_1f=\frac{b}{a_1}-\frac{b_1}{a}=\frac{1}{aa_1}R$) is $=0$.

The coefficient of η is

$$RP\,(cg_1-c_1g)-PQ\,(bg_1-b_1g),$$

which is

$$\begin{aligned}
&= RP\,\frac{1}{bb_1}\,(bc_1 - b_1c_1) - PQ\,\frac{1}{bb_1}\,(b^2 - b_1^2),\\
&= \frac{P}{bb_1}\,[(bc_1 - b_1c_1)\,(ab - a_1b_1) - (b^2 - b_1^2)\,(ca - c_1a_1)],\\
&= \frac{P}{bb_1}\,[b_1^2ca + b^2c_1a_1 - bb_1\,(ac_1 + a_1c)],\\
&= -\,\frac{P}{bb_1}\,\gamma\mathfrak{a}\,;
\end{aligned}$$

and similarly the coefficient of ζ is

$$= -\,\frac{P}{cc_1}\,\mathfrak{a}\beta.$$

We have thus in x, η, ζ the equation

$$-aa_1\gamma x - \frac{P}{bb_1}\,\gamma\mathfrak{a}\,\eta + \frac{P}{cc_1}\,\mathfrak{a}\beta\,\zeta = 0,$$

or, what is the same thing,

$$\frac{x}{P} + \frac{\eta}{bb_1\beta} + \frac{\zeta}{cc_1\gamma} = 0\,;$$

and in like manner, by the elimination of (z, x) and (x, y) respectively,

$$-\frac{\xi}{aa_1\mathfrak{a}} + \frac{y}{Q} + \frac{\zeta}{cc_1\gamma} = 0,$$

$$\frac{\xi}{aa_1\mathfrak{a}} + \frac{\eta}{bb_1\beta} + \frac{z}{R} = 0,$$

which are the required three equations.

We have next to show that the line is a bitangent—write for a moment $aa_1\mathfrak{a}$, $bb_1\beta$, $cc_1\gamma = \lambda$, μ, ν, then the three equations give

$$x,\ y,\ z = P\Big(-\frac{\eta}{\mu} + \frac{\zeta}{\nu}\Big), \quad Q\Big(-\frac{\zeta}{\nu} + \frac{\xi}{\lambda}\Big), \quad R\Big(-\frac{\xi}{\lambda} + \frac{\eta}{\mu}\Big),$$

and we thence have

$$\begin{aligned}\Omega = P^2&\left(-\frac{\xi\eta}{\mu}+\frac{\xi\zeta}{\nu}\right)^2\\ &+Q^2\left(-\frac{\eta\zeta}{\nu}+\frac{\xi\eta}{\lambda}\right)^2\\ &+R^2\left(-\frac{\xi\zeta}{\lambda}+\frac{\eta\zeta}{\mu}\right)^2\\ &-2QR\left(-\frac{\eta\zeta}{\nu}+\frac{\xi\eta}{\lambda}\right)\left(-\frac{\xi\zeta}{\lambda}+\frac{\eta\zeta}{\mu}\right)\\ &-2RP\left(-\frac{\xi\zeta}{\lambda}+\frac{\eta\zeta}{\mu}\right)\left(-\frac{\xi\eta}{\mu}+\frac{\xi\zeta}{\nu}\right)\\ &-2PQ\left(-\frac{\xi\eta}{\mu}+\frac{\xi\zeta}{\nu}\right)\left(-\frac{\eta\zeta}{\nu}+\frac{\xi\eta}{\lambda}\right);\end{aligned}$$